$$(X - Y) = Unexecuted\ Shares\ (Residual)$$

$$\theta = \frac{Y}{X} = Percentage\ of\ Shares\ Traded$$

$$(1 - \theta) = \frac{X - Y}{X} = Percentage\ of\ Shares\ Remaining$$

$$ADV = Average\ Daily\ Volume$$

$$V_t = Volume\ over\ Trading\ Horizon\ (excluding\ the\ order)$$

$$\sigma = Annualized\ Volatility$$

$$\alpha = Trade\ Rate\ at\ Time = t$$

$$POV_t = Percentage\ of\ Volume\ at\ Time = t$$

$$P_t = Market\ Price\ at\ Time = t$$

$$\overline{P}_t = Realized\ Average\ Execution\ Price\ at\ Time = t$$

$$\mu = Natural\ Price\ Appreciation\ of\ the\ Stock\ (not\ caused\ by\ trading\ imbalance)$$

Important Equations

I-Star $\quad I'_{\$/Share} = a_1 \cdot \left(\frac{X}{ADV}\right)^{a2} \cdot \sigma^{a3} \cdot P_0 \cdot 10^{-4} \qquad (8.1)$

Market Impact $\quad MI'_{\$/Share}(\alpha) = b_1 \cdot I' \cdot \alpha + (1 - b_1) \cdot I' \qquad (8.2)$

Timing Risk $\quad TR'_{\$/Share}(\alpha) = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \alpha^{-1}} \cdot P_0 \qquad (8.3)$

Price Appreciation Cost $\quad PA'_{\$/Share}(\alpha) = \frac{X}{ADV} \cdot \frac{1}{\alpha_t} \cdot \mu_t \qquad (8.4)$

Future Price $\quad E(P_n) = P_0 + (1 - b_1) \cdot I' \qquad (8.5)$

Benchmark Cost $\quad Cost = (\overline{P} - P_b) \cdot Side \qquad (8.6)$